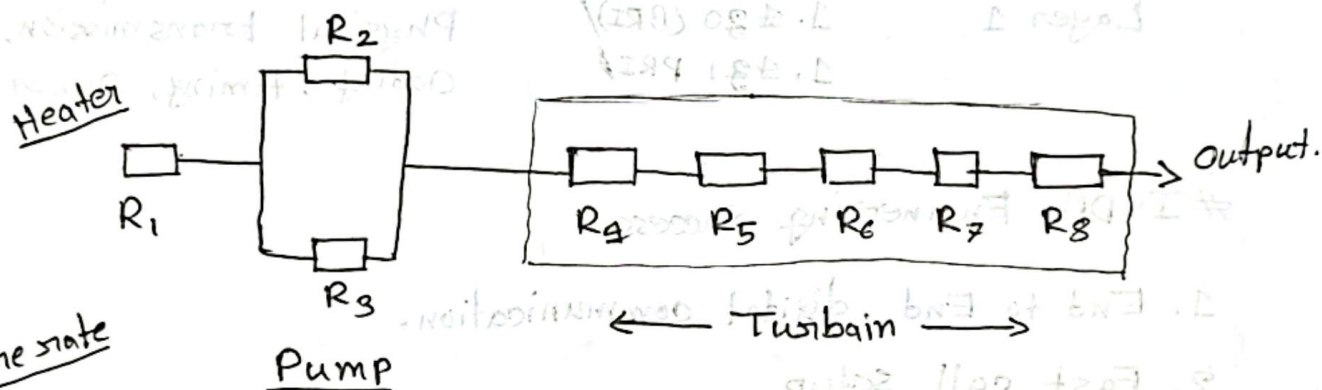


Reliability

Simulation

Turbain Pump Math-

Simple circuit-



Failure rate

$$R_1 = 0.05$$

$$R_2 = 0.10$$

$$R_3 = 0.08$$

$$R_4 = 0.20$$

$$R_5 = 0.17$$

$$R_6 = 0.09$$

$$R_7 = 0.15$$

$$R_8 = 0.15$$

Condition,

Pump - minimum 1 work

Turbain - n 3 n

-o-

Heater reliability, $R_1 = 1 - 0.05 = 0.95$

Pump reliability, $R_{23} = 1 - (0.10 \times 0.08) = 0.92$

For Turbain Reliability,

$$R_4 = 1 - 0.20 = 0.8$$

$$R_5 = 1 - 0.17 = 0.83$$

$$R_6 = 1 - 0.09 = 0.91$$

$$R_7 = 1 - 0.15 = 0.85$$

$$R_8 = 1 - 0.15 = 0.85$$

Counting Table

R_4	R_5	R_6	R_7	R_8	Total Reliability
✓	✓	✓	✓	✓	$0.85 \times 0.8 \times 0.83 \times 0.91 \times 0.85 = 0.435$
✓	✓	✓	✓	✗	$0.8 \times 0.83 \times 0.91 \times 0.85 \times 0.15 = 0.077$
✓	✓	✓	✗	✓	$0.8 \times 0.83 \times 0.91 \times 0.15 \times 0.85 = 0.077$
✓	✓	✗	✓	✓	$0.8 \times 0.83 \times 0.09 \times 0.85 \times 0.85 = 0.043$
✓	✗	✓	✓	✓	$0.8 \times 0.17 \times 0.91 \times 0.85 \times 0.85 = 0.089$
✗	✓	✓	✓	✓	$0.2 \times 0.83 \times 0.91 \times 0.85 \times 0.85 = 0.109$
✗	✗	✗	✓	✓	$0.2 \times 0.17 \times 0.09 \times 0.85 \times 0.85 = 0.022$
✓	✗	✗	✓	✓	$0.8 \times 0.17 \times 0.09 \times 0.85 \times 0.85 = 0.008$

R_4	R_5	R_6	R_7	R_8	Total Reliability
✓	✓	x	x	✓	$0.8 \times 0.83 \times 0.09 \times 0.15 \times 0.85$ $= 0.0076$
✓	✓	✓	x	x	$0.8 \times 0.83 \times 0.21 \times 0.15 \times 0.15$ $= 0.0135$
x	✓	x	✓	✓	$0.2 \times 0.83 \times 0.09 \times 0.85 \times 0.85$ $= 0.0107$
x	✓	✓	x	✓	$0.2 \times 0.83 \times 0.21 \times 0.15 \times 0.85$ $= 0.0122$
x	✓	✓	✓	x	$0.2 \times 0.83 \times 0.21 \times 0.85 \times 0.15$ $= 0.0122$
✓	x	✓	x	✓	$0.8 \times 0.17 \times 0.21 \times 0.15 \times 0.85$ $= 0.0157$
✓	x	✓	✓	x	$0.8 \times 0.17 \times 0.21 \times 0.85 \times 0.15$ $= 0.0157$
✓	✓	x	✓	x	$0.8 \times 0.83 \times 0.09 \times 0.85 \times 0.15$ $= 0.0076$

$$\text{Total} = 0.2707$$

So, total system reliability

$$= \text{Pump Reliability} \times \text{Heater Reliability} \times \text{Turbine Reliability}$$

$$= 0.992 \times 0.95 \times 0.2707$$

$$= 0.2147$$

$$= 21.47\%$$